GrappStar Lean Canvas

2. Problem

- *painful issue and who experiences it*
- *1 or 2 most important difficulties*

Slopes, loose regolith and crevasse edges immobilise ground robots, immediately ending decade long, multi billion dollar missions. Conventional anchor systems winch from existing anchor points, which do not exist on other planets, so one unexpected entrapment has no recovery option.

4. Solution

- *how it meets customer needs*
- *unique process, assembly or technology*

When a rover is stuck or faces impassable terrain, it launches a tethered dart. Live video and IMU feedback guide the dart to self right, then twin augers drive into the regolith. The rover winches free and recovers the dart.

5. Unique Value Proposition

- *why better than all others*
- *why chosen over existing alternatives*

Grappling hooks live in movies, not industry, because Earth offers something to anchor to. Other worlds offer nothing, yet low gravity extends range, so a launched anchor creating its own hold becomes the perfect recovery tool without needing existing infrastructure.

6. Sustainable Advantage

- *start with and maintain differentiation*
- *keep distance from other players*

The design leverages underactuation: a single premium actuator drives multiple degrees of freedom across reel and launcher, exploiting geometry instead of motors. Current practice frees a stuck rover with another rover, risking both. Carrying its own anchor removes that dependency.

7. Customer Segments

- *demographics and interests*
- *what makes them a single market*

Agencies and companies whose rovers must enter terrain they cannot climb out of: NASA's Artemis campaign targeting permanently shadowed craters, ESA's Rosalind Franklin Mars rover, JAXA and ISRO lunar missions, and commercial landers delivering their payloads. All lack an anchor.

3. Existing Alternatives

- *what the audience does today*
- *#1 place they turn*
- *indirect competition*

Stuck rovers today depend on wheel manoeuvres commanded from Earth, which spent eight months failing to free Spirit in 2009. Tethered systems like Dante II instead rely on a crew rigging the anchor at a volcano rim, impossible off Earth.

10. Key Metrics

- *2 to 3 metrics that correlate linearly with success*
- *financial, impact or adoption*

Holding force: prototype 1000N, flight unit 5000N, sufficient to recover a Perseverance class rover (1025kg) from a 30 degree Martian slope requiring 1900N. Range scales inversely with gravity: 10m on Earth becomes 26m on Mars and 60m on the Moon.

1. High-Level Concept

- *simplest, catchiest explanation*
- *impact to target customers*

Ground Retaining Anchor Projectile Placed Self Tensioning Auger Rig

9. Channels

- *attract the market*
- *deliver to customers: partners, internet*
- *how competitors do it today*

Recovery hardware has no buyer until missions dare steep terrain, so we build proof ourselves through terrain trials with university rover laboratories and agency test facilities, supplying directly. When crater missions arrive, rover manufacturers integrate the system as standard equipment.

8. Early Adopters

- *very first customers*
- *specific geography or preexisting connection*

Lunar rovers reach craters within years, before crewed Mars. Small wheeled rovers on commercial landers, such as Astrobotic and Intuitive Machines payloads and university rovers like Carnegie Mellon's Iris, rely on wheels alone and need lightweight recovery: our starting point.

11. Cost Structure

- *recurring variable costs, main drivers*
- *ongoing cost to produce and deliver*

Flight unit: Maxon space grade brushless motors with Harmonic Drive gearheads, Braycote vacuum lubricant, a radiation hardened controller (Microchip SAMRH71), and Vectran tether replacing Dyneema. Then qualification: vibration, thermal vacuum and regolith simulant trials. Launch mass: $1M per lunar kilogram.

12. Revenue Streams

- *ongoing revenue from operations*
- *sales, grants, donations*
- *not upfront funds*

Supply and integration contracts with space agencies and private lander and rover companies carrying public missions, as the sector follows SpaceX's commercial model. Priced on savings: kilograms of anchor against a second recovery rover, or a decade long mission lost.